

Spread Formations

2x2, trips, bunch, empty and the condensed revolution: why offenses spread the field, why they squeeze it back together, and how to name what you see in two seconds.

Third-and-7, early in the fourth quarter. The Bills break the huddle and the running back jogs out wide with the receivers. Josh Allen stands alone in the shotgun: five receivers, three to his right and two to his left, nobody beside him. Before the snap, the Bills' opponent answers. A linebacker walks out of the middle of the field and stands across from the back who just left. A safety creeps down over the slot. Allen points, shouts, and looks at the linebacker one more time. What did the offense learn by standing like that, what did it give up, and why did one defender's walk tell Allen so much? A few seconds before every snap, the offense poses a question with *where its players stand*, and the defense has to answer it in public. This chapter teaches you to read the question.

You'll be able to...

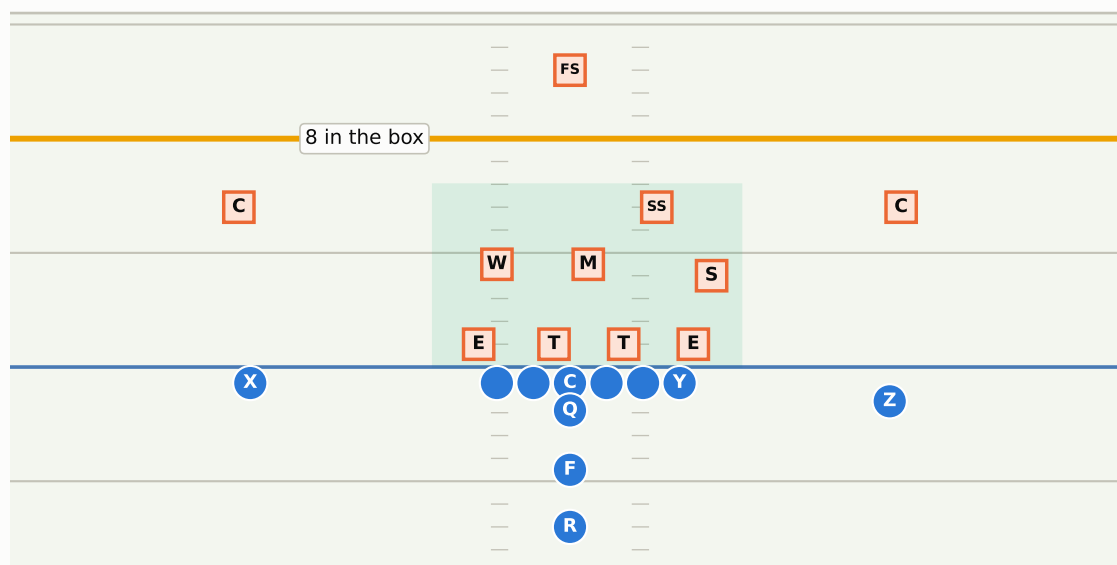
- Name the common spread formations on sight: 2x2 (doubles), 3x1 (trips), empty, bunch, stack, nub and quads, by counting receivers on each side of the ball.
- Explain why spreading receivers across the field forces the defense to show what it is doing before the snap, and how it thins the defense in the middle of the field for the run game.
- Do "the count": receivers against defenders on each side, and say which side the offense is likely attacking.
- Explain why modern NFL offenses squeeze their receivers back toward the ball (condensed splits), and name three things a tight split buys: crack blocks, a two-way go, and rubs.
- Recognize the specialty looks (wildcat, unbalanced line, swinging gate) and say what each one is trying to trick the defense into.

From tight to spread: why width is a weapon

Classic Formations was a story about bodies near the ball: a fullback leading the way, two tight ends, the I-formation's straight line of backs. Those formations put blockers where the run happens. The trouble is that they also invite the defense to put *its* bodies there. If every offensive player is within a few yards of the ball, every defender can be too.

A spread formation flips that. The offense takes the same eleven players and stretches them across the field's full width: 53⅓ yards, sideline to sideline. A receiver standing near the sideline is a threat to catch the ball there, so someone has to stand near him. Every defender who walks out to cover a receiver is a defender who is *not* standing near the ball waiting for the run.

Tight: I-formation, 21 personnel



Spread: 2x2, 11 personnel

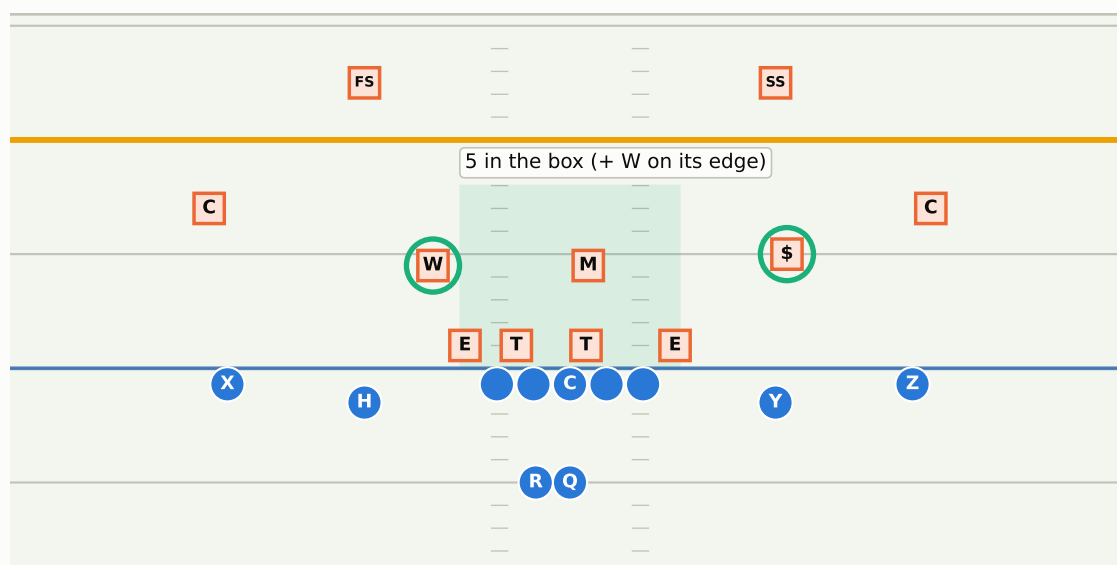


Figure 1: The same eleven defenders against two ways of using the field. Top: 21 personnel in an I-formation, every offensive player within a few yards of the ball, and the defense answers with eight in the box (shaded): four linemen, three linebackers and a safety. Bottom: 11 personnel spread out 2x2 from the shotgun. Four receivers stretch the field, and the defense has to send players with them: the nickel back (\$) to one slot, the Will linebacker (W) halfway out toward the other, two corners and two deep safeties. Five defenders are left in the box, with a sixth on its edge.

That picture is the first half of the spread idea: **width empties the box**. The **box** is the area near the ball, roughly from tackle to tackle and five to seven yards deep, where run defenders stand; the number of defenders in it is the single most important clue for whether a run will work. (The box gets a full treatment in A First Look at Defense, the next chapter after this one.) Two fewer defenders near the ball means two fewer people to block on a run.

The second half of the idea is subtler, and it's the one coaches talk about most: **width makes the defense declare**. When receivers are tight to the formation, a defender can stand in the middle of everything and decide after the snap whether to cover, rush or play the run. When receivers spread out, defenders have to walk to them *before the snap*, and where they walk tells the quarterback a great deal. Does a cornerback line up right on top of his receiver, or eight yards off? Does a linebacker go out with the back who split wide, or does a safety? Is there one safety deep in the middle of the field, or two? Each of those choices is a clue to the coverage, and spread formations force the defense to make them in plain sight. In Motion and Shifts as Weapons you'll see offenses push this further by moving players around before the snap and watching who follows.

! Why it exists

Spread formations solve two problems at once. **Run problem:** too many defenders near the ball. Spreading receivers pulls defenders out of the box. **Pass problem:** not knowing the coverage until after the snap. Spreading receivers forces defenders to align to them, so the quarterback gets clues before the ball is snapped. **The trade:** every player split wide is a player who isn't blocking. A spread offense has fewer blockers near the ball for runs and for protecting the quarterback, which is why the quarterback's ability to throw quickly, or run himself, matters so much more in the spread.

Where the spread came from

The NFL didn't invent the spread. It borrowed it, decades late, from high schools and colleges. A short version (the full story is in The College Laboratory):

- In the 1950s, Glenn "Tiger" Ellison, a high-school coach in Middletown, Ohio, was losing with a conventional offense and started spreading his receivers and letting them adjust their routes to the coverage they saw. He wrote it up as *Run and Shoot Football: Offense of the Future* in 1965.¹ Mouse Davis carried the **run and shoot** into college and then the NFL, where the Houston Oilers and Detroit Lions ran four-receiver versions of it in the early 1990s.
- In 1989 Hal Mumme and his young assistant Mike Leach took over at tiny Iowa Wesleyan and built what became the **Air Raid**: four or five receivers spread wide, the same handful of pass plays repeated until they were perfect.
- Around the same time, coaches such as Rich Rodriguez and Urban Meyer spread the field to *run* the ball, with the quarterback reading a defender instead of handing off. That family is the **spread option** (Option Football).

What these coaches shared was a problem: they usually had smaller, slower players than their opponents. Lining up tight and running into a bigger defense was a losing fight. Spreading out turned the game into one of space and arithmetic, and in space a good athlete beats a big one.² The NFL took longer to adopt it because NFL teams *had* the big, strong players, and because pro quarterbacks were raised under center. By the 2000s, college quarterbacks were arriving in the league having run the spread their whole careers, and NFL offenses followed them.

i Film room: Patriots 2007

The 2007 New England Patriots went 16-0 in the regular season and scored 589 points, then an NFL record. Tom Brady threw 50 touchdown passes and Randy Moss caught 23, both records at the time, and slot receiver Wes Welker caught 112 passes.³ The way they did it was new for the NFL. Under coordinator Josh McDaniels, they took **50% of their snaps from the shotgun**, the most in the league, when the league average was 27% (our count from nflverse play-by-play, regular season runs and passes). Typical looks: Moss alone on one side, Welker and another receiver on the other, and often nobody in the backfield beside Brady at all.

What to notice: watch how the defenses had to choose. Put a safety over the top of Moss and Welker was alone in the slot against a linebacker or a nickel back with no help; stay balanced and Moss was one-on-one deep. A spread formation hands the defense a list of problems and asks it to choose which one to live with.

Reading the notation: counting receivers

Spread formations have a shorthand that you'll hear on broadcasts and read in every analytics article, and it is just counting. Look at the receivers, meaning everyone who could catch a pass *except* the quarterback and the backs standing in the backfield. Count how many stand to the right of the ball and how many to the left. Write the bigger number first.

- Two and two: **2x2** (say "two-by-two"; coaches also call it **doubles**).
- Three and one: **3x1**, usually called **trips**.
- Three and two, with nobody left in the backfield: **3x2**, which is **empty**.

¹Glenn Ellison, *Run and Shoot Football: Offense of the Future* (West Nyack, NY: Parker Publishing, 1965). Ellison coached at Middletown High School in Ohio; see also Layden, *Blood, Sweat and Chalk*, ch. on the run and shoot.

²Tim Layden, *Blood, Sweat and Chalk: The Ultimate Football Playbook* (New York: Sports Illustrated Books, 2010), chapters on the run and shoot, the Air Raid and the spread option. For the Air Raid's start at Iowa Wesleyan in 1989, see S. C. Gwynne, *The Perfect Pass* (New York: Scribner, 2016).

³Pro Football Reference, 2007 New England Patriots: <https://www.pro-football-reference.com/teams/nwe/2007.htm>. The 589 points were broken by the 2013 Denver Broncos (606). Shotgun shares are this chapter's count from nflverse play-by-play (shotgun flag), 2007 regular season, runs and passes.

A tight end on the line counts as a receiver on his side (he is one: see Formation Rules on who is eligible). A fullback or running back in the backfield does not. That's why the count tells you about the passing game: it says how many targets each side of the defense must handle the instant the ball is snapped. You'll also hear **2x1** for a two-back set with a tight end, and **1x1** for the heaviest looks, but most NFL snaps are 2x2 or 3x1.

Inside a side, defenders count too, and they count from the sideline in. The widest receiver is #1, the next one in is #2, the next #3. "Who has #3?" is one of the most common questions a defensive coach asks, and you'll meet the numbering again in Disguise, Rotation, and Split-Field Coverage.

Here is the family on one page. Each plate is drawn with the ball in the middle of the field and the formation's strength to the offense's right; the receivers that make each formation what it is are ringed.

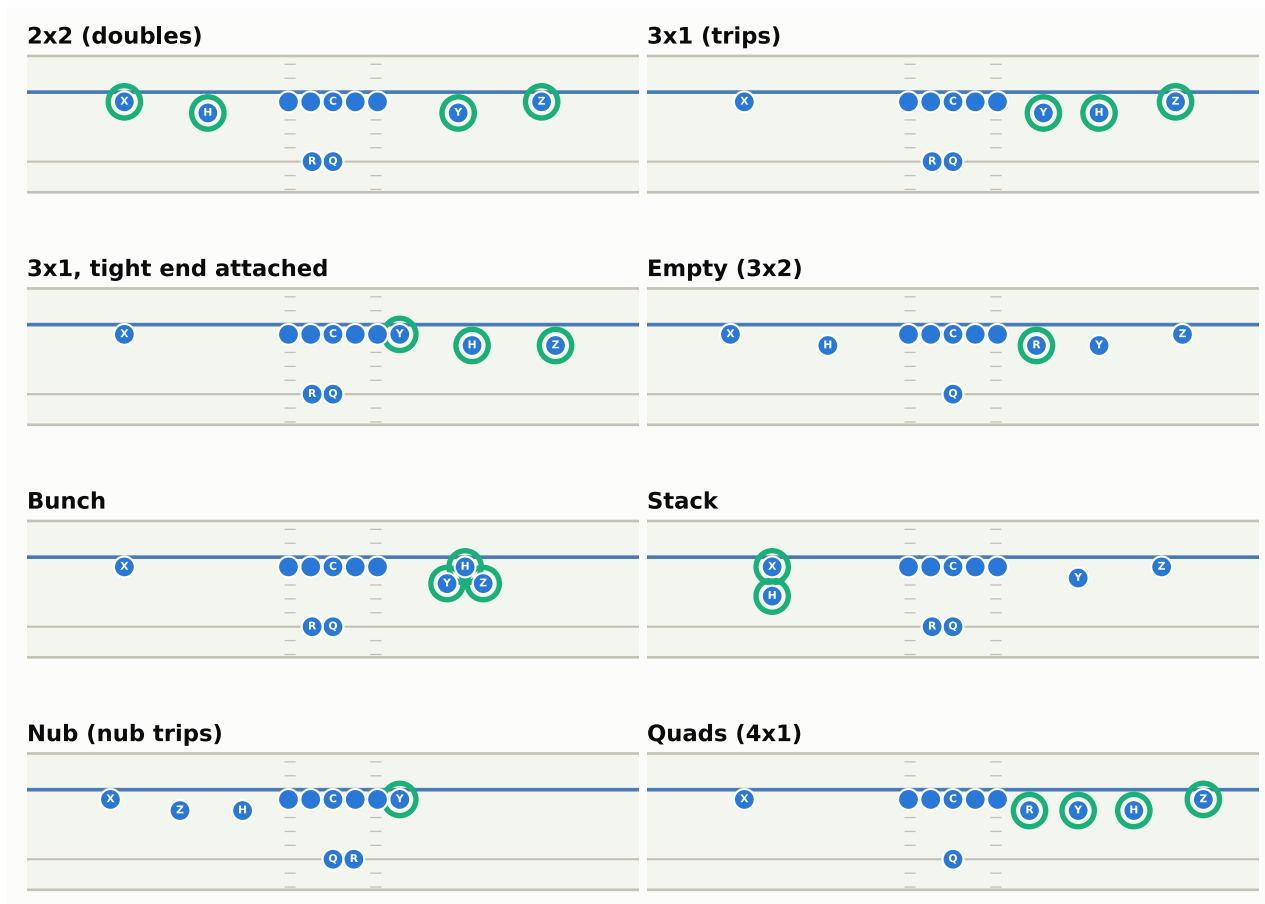


Figure 2: The spread family, offense only, ball in the middle of the field. Ringed players make the formation what it is. 2x2: two receivers each side. 3x1 (trips): three to one side, either all detached or with the tight end (Y) attached to the line. Empty (3x2): the back (R) has left the backfield. Bunch: three receivers in a tight triangle. Stack: one receiver directly behind another. Nub: the tight end is the last man on his side, with no wide receiver outside him (here 'nub trips', three receivers away from him). Quads: four to one side. Every plate has exactly seven players on the line of scrimmage.

i Madden vs. real life

In *Madden* you choose a formation from a menu: "Gun Trips TE," "Gun Bunch," "Gun Empty Base." Real teams don't use shared names. Every team has its own words: one staff's "Trey Right" is another's "Trips Right Y-Attached," and a third team might call the same alignment by a color or an animal. What *is* shared is the description, the counting language in this chapter. Coaches visiting another team, analysts and scouts all say "3x1 with the tight end attached," because that describes where the players stand no matter what the play sheet calls it.

2x2: the balanced spread

2x2, or **doubles**, puts two receivers on each side. In the NFL's most common grouping, 11 personnel (one back, one tight end, three wide receivers), it usually looks like the plate above: two wide receivers on the outside, a slot receiver on one side, and the tight end flexed out as the slot on the other.

Why use it? Because it's **balanced**. Nothing about it tells the defense which way the play is going, and it doesn't let the defense load up one side. Each side has two receivers, so each side needs about the same number of defenders. That makes 2x2 a good "neutral" look: you can run any play from it, and the defense learns little from the formation itself. It also spreads the defense evenly, which keeps the box light.

The weakness of 2x2 is the flip side of its strength: balance lets the *defense* stay balanced too. Two receivers against a cornerback, a slot defender and a safety on each side is a fair fight for the defense, and the coverage doesn't have to twist itself to match. That's the moment to learn the habit this chapter is really about: **the count**.

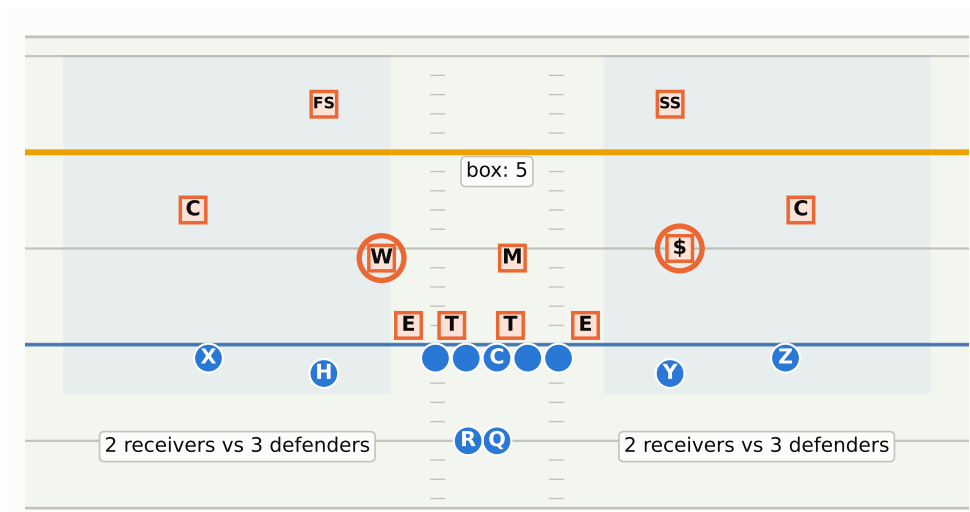


Figure 3: The count against a 2x2. Each side has two receivers, and the defense has three players who can cover them: a cornerback, a slot defender (the nickel back \$ on the right, the Will linebacker W on the left) and a deep safety. Two receivers against three defenders on each side means the defense isn't outnumbered anywhere, but it took three defenders per side, so only five remain near the ball against the offense's five linemen and running back.

To do the count, split the field at the ball. On each side, count receivers, then count the defenders who could cover them: everyone outside the box, plus any linebacker standing halfway out. Compare. If one side has more receivers than defenders, the offense has a numbers advantage there and will often throw to that side, especially a quick throw before the defense can rotate help over. If the defense matches or outnumbers the receivers everywhere, look at the box: those extra defenders came from somewhere, and a light box invites a run. You'll get better at this all season; for now, getting the two numbers on each side is enough.

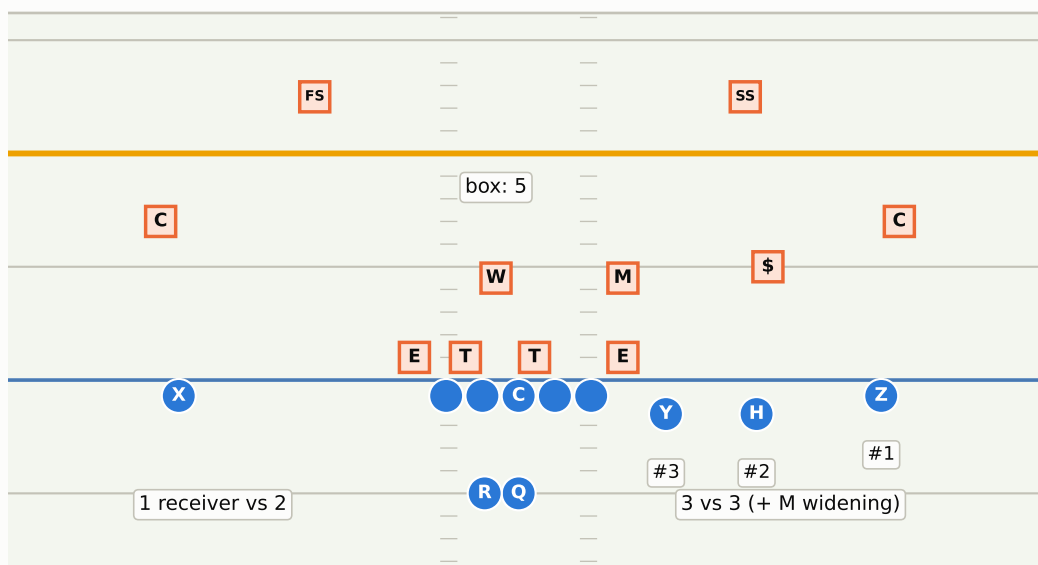
3x1 (trips): the overload

3x1, or **trips**, puts three receivers on one side and leaves one receiver alone on the other. The three can all stand off the line as wide receivers and slots, or the tight end can stay attached to the line as the third. Trips is the most important spread formation for a viewer to recognize, because it forces the defense into a real choice, and the choice is visible.

Here's the problem it creates. The trips side has three receivers. If the defense stays balanced, with its usual two deep safeties and a cornerback on each side, it has a cornerback, a slot defender and a safety on the trips side: three against three, with the safety far back. Someone has to handle the third receiver (#3), the one closest to the ball, and that usually means a linebacker walking out of the box. Meanwhile, on the other side, a cornerback *and* a safety stand guard over a single receiver. That's two defenders for one receiver on one side, and three for three on the other, with the box thinned to five.

The defense's other choice is to **rotate**: move a safety toward the trips side so there are plenty of defenders over there. But the safety had to come from somewhere, and now the lone receiver on the other side (in this book's labels, the X receiver) is alone against one cornerback, with no safety over the top to help. That is exactly what an offense wants when its best receiver is X.

Stay balanced (two-high)



Rotate to trips (one-high)

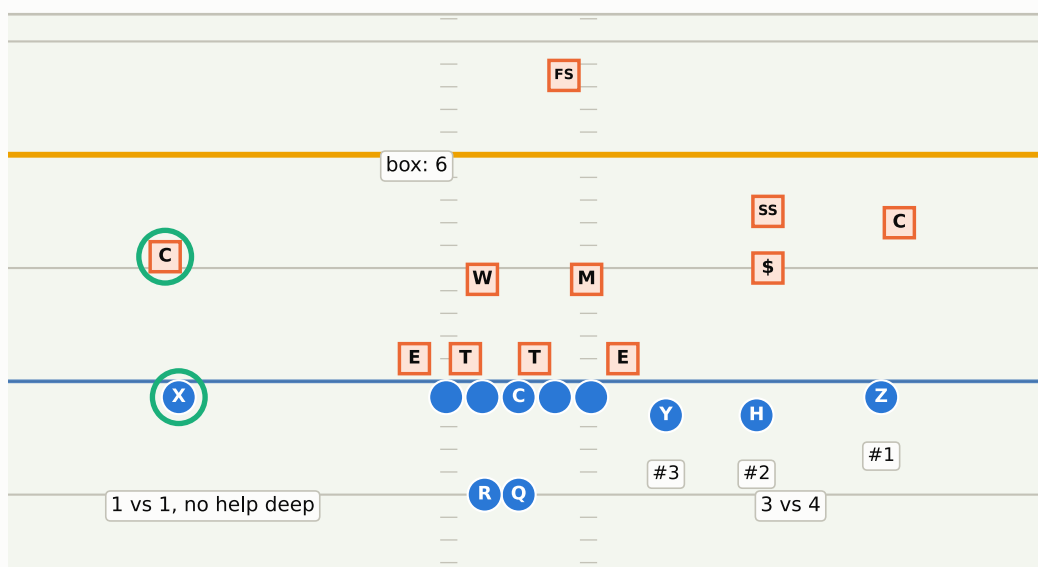


Figure 4: The trips problem, in two pictures. Top: the defense stays two-high and balanced. The trips side is three receivers (#1, #2, #3, counted from the sideline in) against a cornerback, the nickel back and a safety, so the Mike linebacker (M) has to widen toward #3, leaving five in the box. On the left, a cornerback and a safety guard one receiver. Bottom: the defense rotates. The left safety moves to the middle of the field and the right safety drops down toward the trips side. Now there's help everywhere on the trips side, but X, alone on the left, is one-on-one with his cornerback (ringed) and nobody is deep over him.

Neither answer is wrong; each has a cost, and that's the point. Trips makes the defense pick which problem to live with, and the pick shows. When you see trips on Sunday, look at the safeties before the snap: two deep and level means the defense is staying balanced (watch for a run, or a throw to the trips side); one safety sliding toward the middle or the trips side means it is rotating (watch X). Defenses know you are watching, so they disguise this, holding two safeties deep until the snap and rotating at the last instant, and they have special rules for trips that move a safety across without leaving X alone. Those tricks are the subject of Disguise, Rotation, and Split-Field Coverage.

! Why it exists: trips

Problem: a balanced defense is hard to attack because it is never outnumbered. **Answer:** overload one side so the defense must either move players (and leave someone alone on the other side) or stay put (and be outnumbered or thin in the box). **The trade:** the backside receiver is isolated, which is great only if he can win one-on-one, and the formation does tell the defense which side has more targets. Offenses often pair trips with their best receiver as the lone X, precisely so the defense's rotation puts him one-on-one.

Empty: everyone out

Empty sends the running back out of the backfield to join the receivers, so five receivers spread across the field and the quarterback stands alone in the shotgun. The usual version is **3x2**: three to one side, two to the other. (A 4x1 version is called quads, coming up below.)

Empty is the purest form of the spread idea, for both halves of it.

1. **It forces the defense to declare.** Five receivers need five defenders near them. A linebacker who would normally stand in the middle now has to walk out over the running back. If he does, the quarterback has learned something (a linebacker covering a back in space is a mismatch he may want). If a defensive back goes out instead, he has learned something else (the defense has extra defensive backs on the field, and somebody left the box). Either way, man and zone coverage tend to show themselves more clearly against empty, because every receiver needs an answer.
2. **It empties the box.** With the linebackers pulled out to the receivers, there may be only four defenders near the ball: the four who are about to rush the passer.

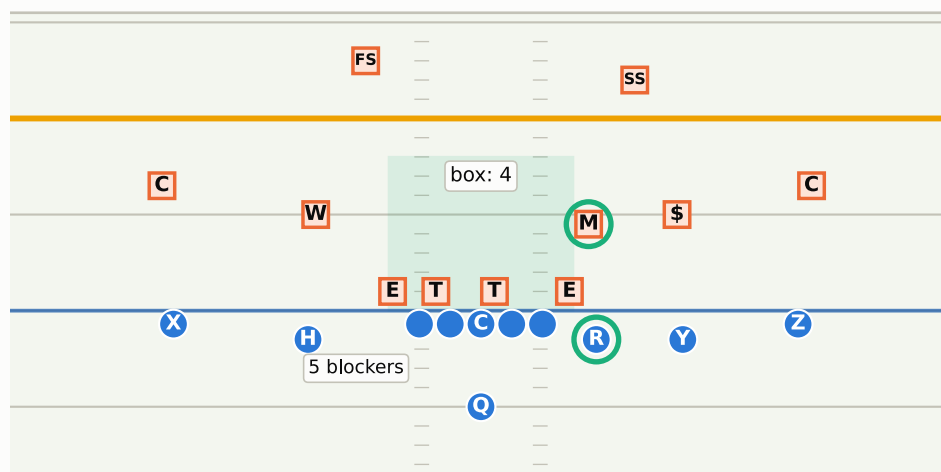


Figure 5: Empty (3x2) against a nickel defense. Every receiver needs someone near him, so both linebackers leave the middle: the Will (W) widens over H on the left and the Mike (M) walks out over the back (R) on the right. The box is down to the four linemen. The ringed Mike is the tell the quarterback looks for: a linebacker lined up across from a running back in space is a match-up the offense may want to throw at.

The cost is just as plain: **nobody stays home to block**. Five offensive linemen have to protect the quarterback against however many defenders rush. If the defense sends six, one rusher is unblocked by design, and the quarterback has to see it and throw before that rusher arrives, usually to a quick route or a **hot route** (a pre-planned quick throw to beat an extra rusher; see Pass Protection). Empty is a bet that the quarterback can process quickly and get the ball out.

That bet shows in the data. The figure below counts the backs in the backfield on every NFL run and pass from 2022 to 2025, using FTN's charting, which nflverse distributes.

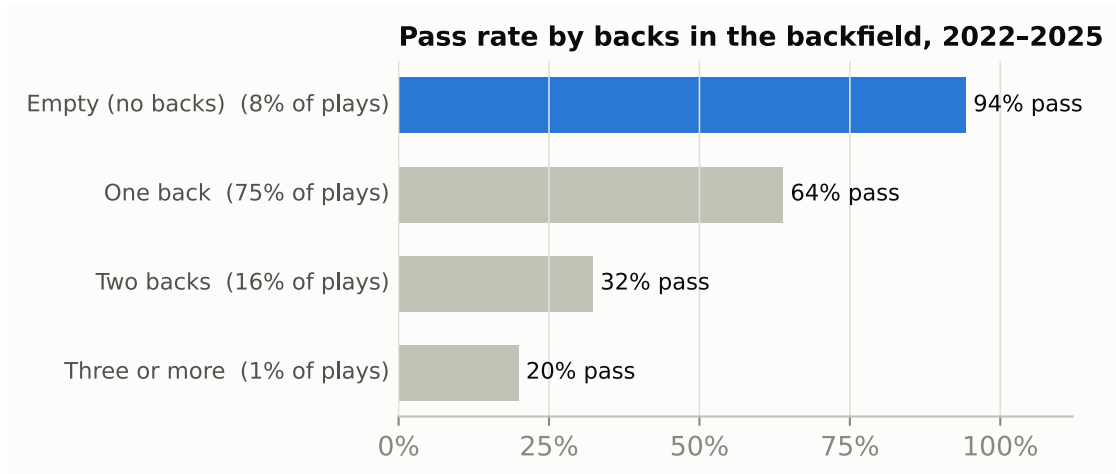


Figure 6: How often the offense passed, by the number of backs standing in the backfield at the snap (the quarterback not counted), all regular-season runs and passes, 2022–2025. Pass plays include sacks and quarterback scrambles, because the call was a pass. From an empty backfield, offenses passed about 94% of the time. Source: nflverse play-by-play joined to FTN charting (n_offense_backfield).

Empty is one of the strongest pass tells in football: about nineteen times in twenty, an empty backfield means a pass (and many of the exceptions are quarterback runs). The defense knows that too, which is why offenses use it sparingly, roughly one snap in twelve, and more on third down, when a pass is expected anyway.

i Film room: Bills 2019–2025, empty with Josh Allen

Buffalo has spent most of the Josh Allen era near the top of the league in empty formations, under three different play-callers (Brian Daboll through 2021, Ken Dorsey in 2022–23, Joe Brady in 2024–25). The table counts it.

Table 1: Share of the Bills’ regular-season runs and passes from an empty backfield, and their rank among the 32 teams. 2019–2022 use the formation label in nflverse participation data (NGS source); 2023–2025 use FTN’s count of backs in the backfield. The two sources measure empty slightly differently (in 2022, when both exist, they agree at about 8% league-wide), so compare ranks across the break, not exact percentages.

Season	Bills empty share	Bills rank (of 32)	League average	Source
2019	12.5%	3	8.3%	NGS label
2020	11.1%	10	9.2%	NGS label
2021	6.8%	26	9.4%	NGS label
2022	11.1%	5	8.1%	NGS label
2023	10.5%	5	7.7%	FTN backs
2024	10.3%	8	8.6%	FTN backs
2025	8.5%	9	6.9%	FTN backs

What to notice: empty is supposed to be a pure pass tell, but Allen bends it. He is a 6-foot-5 quarterback who runs like a fullback, so when Buffalo goes empty, the defense can’t assume the ball will be thrown. From 2022 to 2025, by FTN’s count, the Bills ran a designed run (not a scramble) on about 9% of their empty snaps, sixth-most in the league, against a league average of about 6%, and almost all of those runs were Allen keeping it himself. That small threat changes the defense’s math. If every defender drops into coverage against five receivers, a quarterback draw goes for big yards; if a linebacker stays home to spy Allen, one receiver has less help. The Ravens with Lamar Jackson ran from empty even more often.

Watch a Bills third down: when the back flexes out and Allen is alone, look at the middle linebacker. Does he run out with the back, sink into the middle, or creep toward the line to watch Allen? Whatever he does, Allen has seen it before the snap.

Bunch and stack: squeezing receivers together

So far, every formation has spread receivers *apart*. **Bunch** and **stack** do something that seems backwards: they pile receivers on top of each other.

A **bunch** is three receivers in a tight triangle, usually one on the line of scrimmage (the “point”) and two a yard or so behind him, one to each side. A **stack** is two receivers lined up one behind the other, a couple of yards apart. Both are usually set a little closer to the ball than a normal wide receiver.

Why squeeze them together? Three reasons, all about what happens in the first two seconds.

1. **They beat press coverage.** A cornerback who lines up right on top of his receiver to jam him at the line (**press**, taught in Coverage Fundamentals) can only press one man. In a stack or bunch, the receivers behind the point man can’t be jammed, because there’s a teammate in the way. They get a free release.
2. **They hide who’s going where.** Before the snap, the defense sees a cluster, not three separate receivers with obvious paths. Any of them could go inside, outside, short or deep.
3. **They create traffic.** When three receivers start from nearly the same spot and run in different directions, their paths cross. Defenders chasing them in man coverage have to fight through the crowd, and one of them gets caught behind it. Plays designed to make that happen are called **rub routes** (or pick routes), and they get their full treatment in Dropback Concepts.

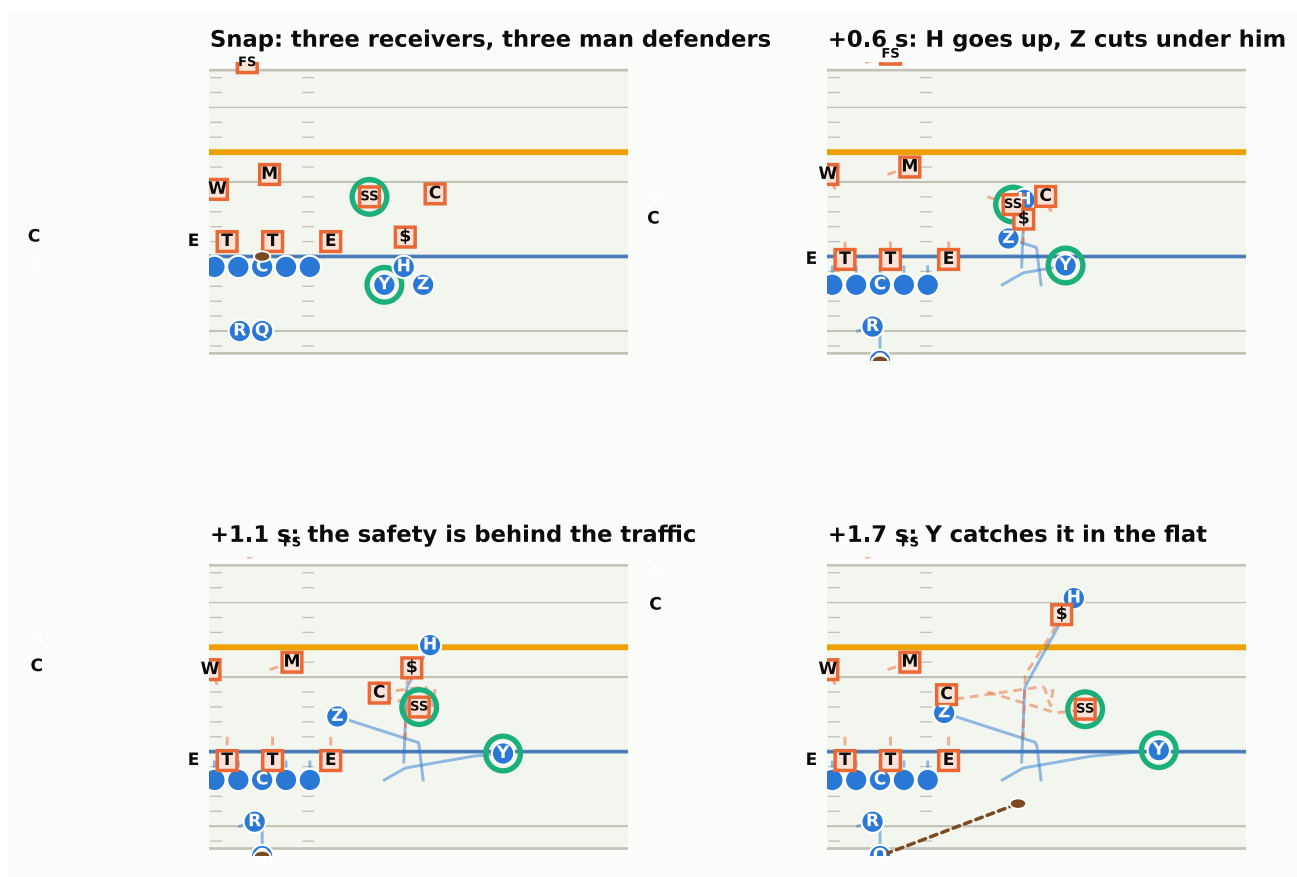


Figure 7: Bunch against man coverage, simulated. Three defenders each take one receiver in the bunch (the nickel back on the point man H, the right corner on Z, the strong safety on Y, the tight end). At the snap H runs straight upfield and Z cuts inside underneath him, while Y, starting from the inside, runs to the flat. The safety chasing Y has to get through H and Z and their two defenders, so Y is open in the flat. In print, the brown dashes are the ball in the air.

There’s a rule lurking here. A receiver isn’t allowed to deliberately block a defender while a pass is in the air, and the officials call a deliberate pick downfield offensive pass interference. Good bunch plays live right at the edge of that rule: the receivers run *routes* that happen to cross, close to the line of scrimmage, and any contact looks like two players in the same place at the same time.⁴ That’s one reason bunch concepts are so often quick throws.

⁴NFL rulebook, Rule 8, Section 5 (pass interference), which lists “blocking downfield” by an offensive player before the pass is touched as offensive pass interference: <https://operations.nfl.com/the-rules/nfl-rulebook/>.

Defenses don't play bunch with simple man-to-man for long. The usual answer is to stop assigning a defender to a *person* and assign him to a *direction*: whoever releases outside first belongs to the outside defender, whoever comes inside belongs to the inside defender. That's called a **banjo** or switch call, and you'll meet it in Man Coverage. Zone defenses handle bunch more naturally, which is part of why the offense uses it: if a bunch makes the defense play zone, the offense has learned something.

A **stack** works on the same ideas with two receivers instead of three. The front man takes the cornerback's attention (and his jam, if the corner presses), and the back man gets a free release and a choice of directions: he can break outside behind the front man or inside underneath him.

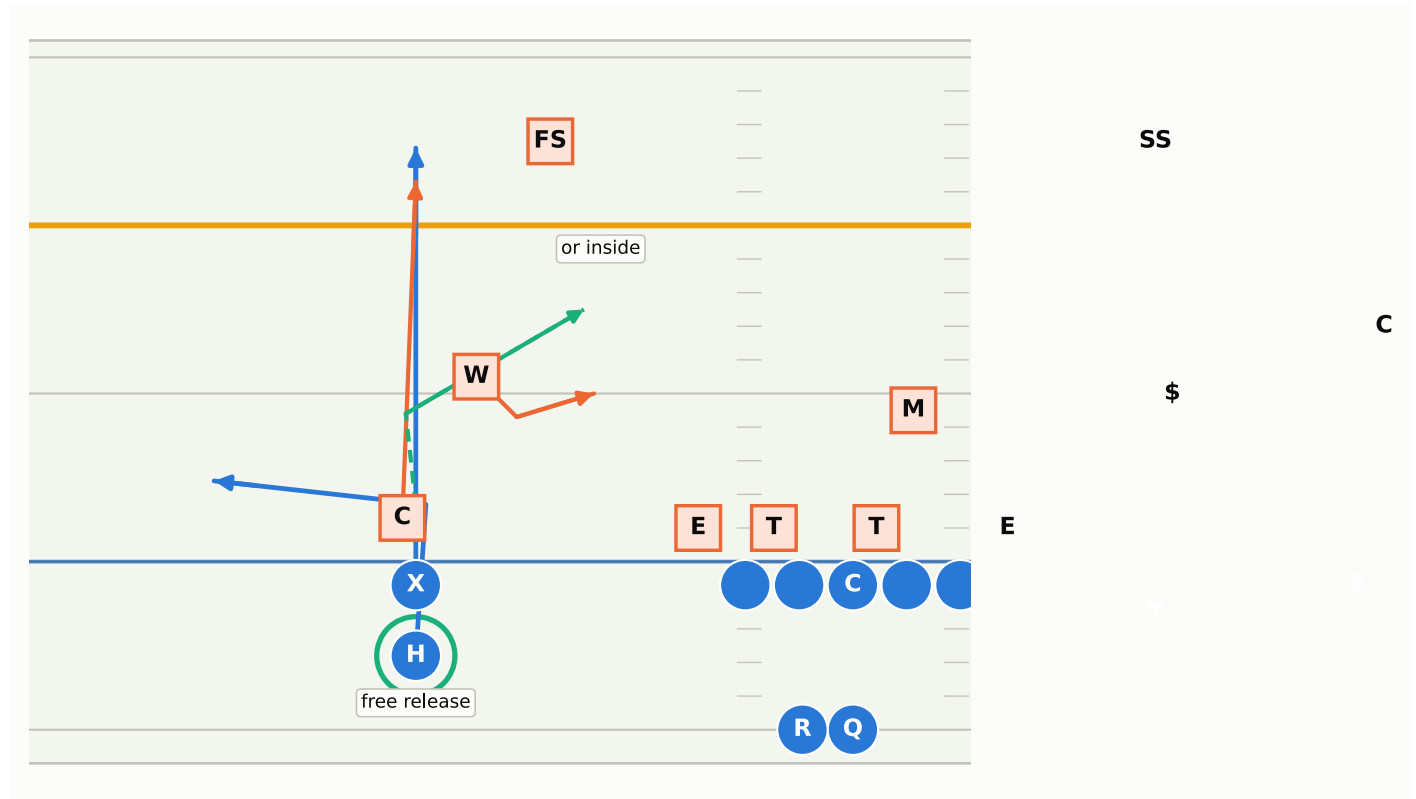


Figure 8: A stack on the offense's left side against press-man coverage. The cornerback can jam only the front receiver (X). X runs straight upfield and takes the corner with him; H, the back receiver, gets a free release and breaks out to the sideline underneath, while his defender, lined up behind the corner, has to run around both of them. H could just as easily break inside (dashed), and the defender can't know which until H commits.

i Film room: Chiefs bunch and stack sets under Andy Reid

Andy Reid's Kansas City offense has leaned on bunch and stack sets for years, especially near the goal line and on third down, often with Travis Kelce as one of the three in the bunch. The reasoning is the one above. On a short field the defense has less room to play off its receivers and more reason to press and play man, which is exactly what bunch punishes. And Kelce, a tight end who is bigger than a defensive back and faster than most linebackers, gives the defense no easy man-to-man match-up even before the traffic starts.

What to notice: on a Chiefs third-and-short, when three receivers huddle together a step apart, watch the defenders over them before the snap. If they line up one-on-one, each staring at his own man, expect the bunch to cross and someone to come open underneath. If they stand at different depths and point at each other, they've called a banjo, and the offense will have to win another way. (More in the Chiefs case study.)

Nub and quads: the two extremes

Two more names complete the family, one about what's *missing* from a side and one about what's piled on it.

A **nub** side is a side with a tight end on the line and *nothing* outside him: no wide receiver. "Nub trips" is the classic version: three receivers to one side, and on the other the tight end alone at the end of the line. Why would you leave a side so bare? Because the

tight end there is a blocker on the edge of the line *and* a receiver, and because the defense's cornerback on that side has nobody to cover outside him. The corner can join the run defense (which is what the offense wants him to think about) or rotate away to help against the trips side (which leaves the tight end with room to run a route into). Nub sides also squeeze the field: with no receiver near the sideline, the defense's edge players can pinch inward, and that's where outside runs and bootlegs, the quarterback rolling away from the flow of the play, go to work.

Quads puts four receivers on one side. It is almost always an empty formation (4x1), because the rules of the line of scrimmage make it awkward otherwise: you need seven players on the line, and with four receivers on one side you need someone on the line on the other side too. Quads is the trips problem made worse: the defense now needs four defenders to that side, and the lone receiver on the other side is even more alone.

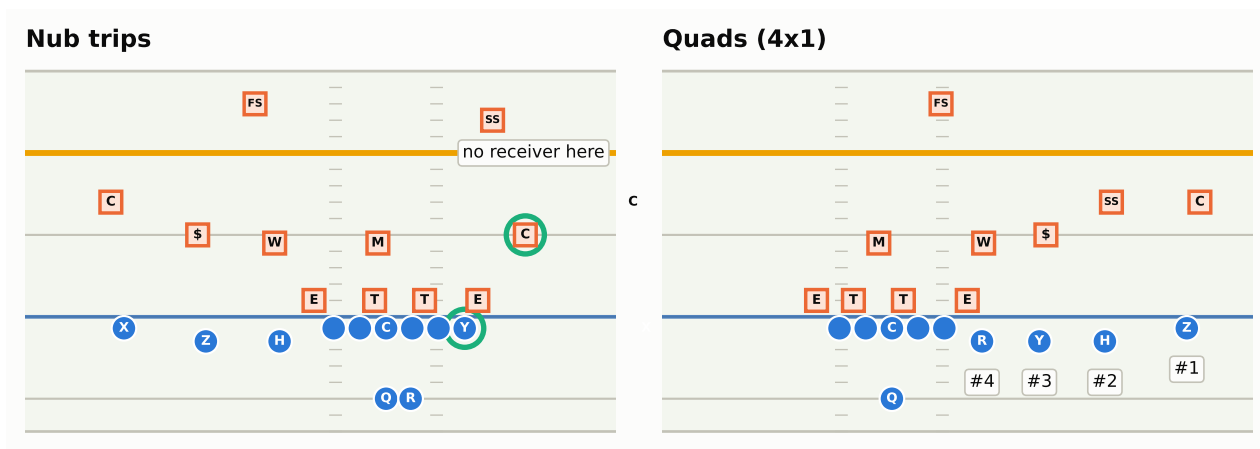


Figure 9: Two extremes. Left: nub trips. The tight end (Y, ringed) is the last man on the right side of the line, with no receiver outside him, so the right cornerback (ringed) has no one to cover outside and can creep in toward the run or rotate away. Three receivers stand on the other side. Right: quads (4x1 empty). Four receivers to one side need four defenders; the box shrinks to the four linemen plus one linebacker.

The condensed revolution

Everything so far says width is good. So here's the twist that defines the modern NFL: over the last decade, many of the league's best offenses have pulled their receivers *in*. Instead of lining up near the numbers, a receiver stands four or five yards outside the tackle or tight end, in a reduced or tight split (the split ladder is in Formation Rules). A formation built this way is a **condensed formation**.

It looks like a step backward, toward the tight formations of the past. It isn't, because the receivers are still receivers; they've just moved to a spot where they can do more jobs. A condensed receiver can do at least four things a receiver on the numbers can't.

1. Room to the outside

Start with geometry. A receiver near the numbers has about a dozen yards between him and the sideline. A cornerback covering him can treat the sideline as a twelfth defender: he lines up inside the receiver, takes away everything toward the middle, and dares him to beat him toward a sideline that's coming up fast. Pull the same receiver in toward the ball and the sideline is twenty yards away. Now an out-breaking route has a whole field to land in.

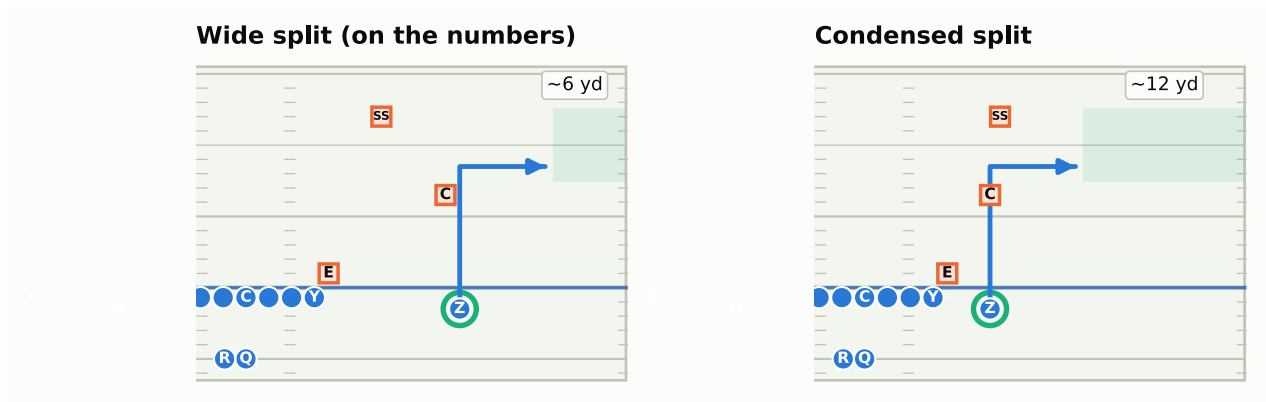


Figure 10: The same route from two splits, right half of the field, ball in the middle. Left: Z lines up on the numbers, and his 10-yard out lands with about 6 yards to spare before the sideline; the corner back can play inside him and let the sideline help. Right: Z lines up in a condensed split, about four yards outside the tight end. The same out route now has about 13 yards of room to the outside (shaded), and the corner back can't lean on the sideline.

2. A two-way go

Room on both sides creates the most valuable thing a receiver can have: a **two-way go**. A defender covering a receiver normally takes away one direction by lining up on that side of him (inside or outside **leverage**, which The Quick Game teaches in full) and relies on the sideline or a teammate to cover the other. A condensed receiver has space inside *and* outside, with no sideline nearby. If the defender lines up inside, the receiver breaks out; if he lines up outside, the receiver breaks in; if he lines up head-up, he has to guess. The receiver gets to choose after he sees which way the defender is leaning.

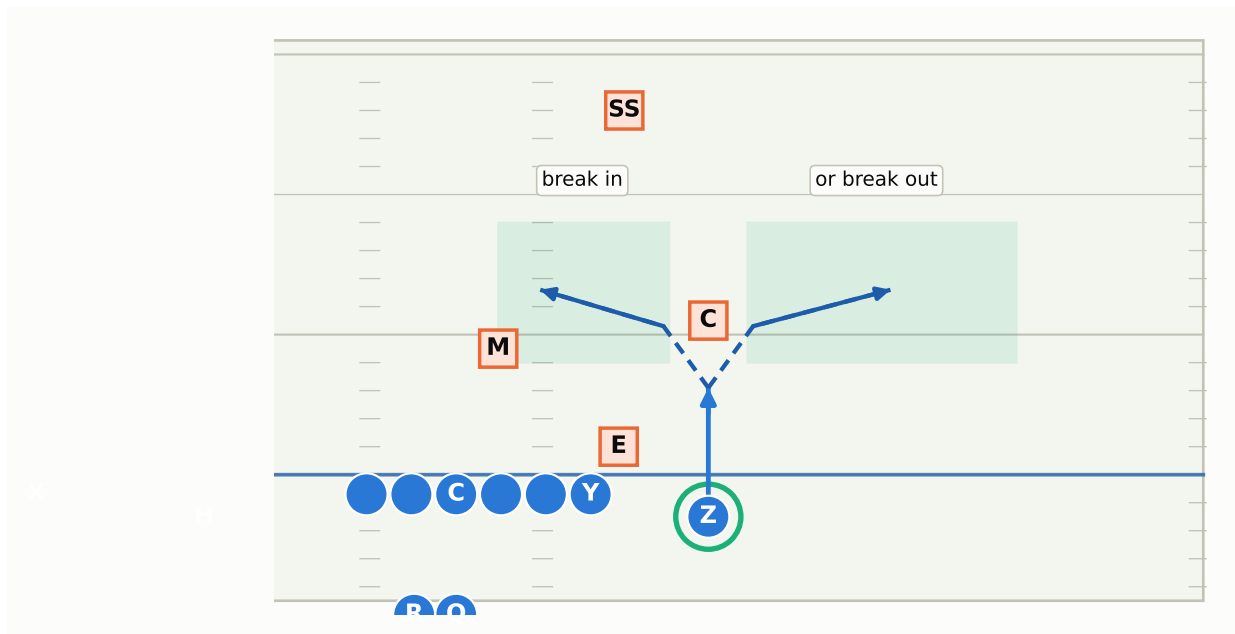


Figure 11: A two-way go. Z lines up in a condensed split and the defender over him plays him head-up, with no sideline or teammate close enough to help on either side. Z pushes upfield five yards, then breaks whichever way the defender doesn't take away: inside toward the open middle (left branch) or outside toward the wide-open sideline (right branch). Both shaded areas are his.

3. Rubs and crossing routes

Pull two or three receivers in close to each other and you get bunch-like traffic without a bunch. Receivers who start close together can cross each other's paths quickly, which makes man coverage miserable for the defense, for the reasons in the bunch section.

4. Crack blocks: a receiver who can block like a tight end

This is the one that changed the NFL's run game. From a condensed split, a receiver is close enough to the box to block a linebacker or safety *inside* him. Blocking back toward the ball on a defender who's looking the other way is called a **crack block** (taught

fully in Perimeter Runs and Misdirection). From out on the numbers, the receiver is too far away to reach those defenders before the play is over. From a tight split, he's two steps away.

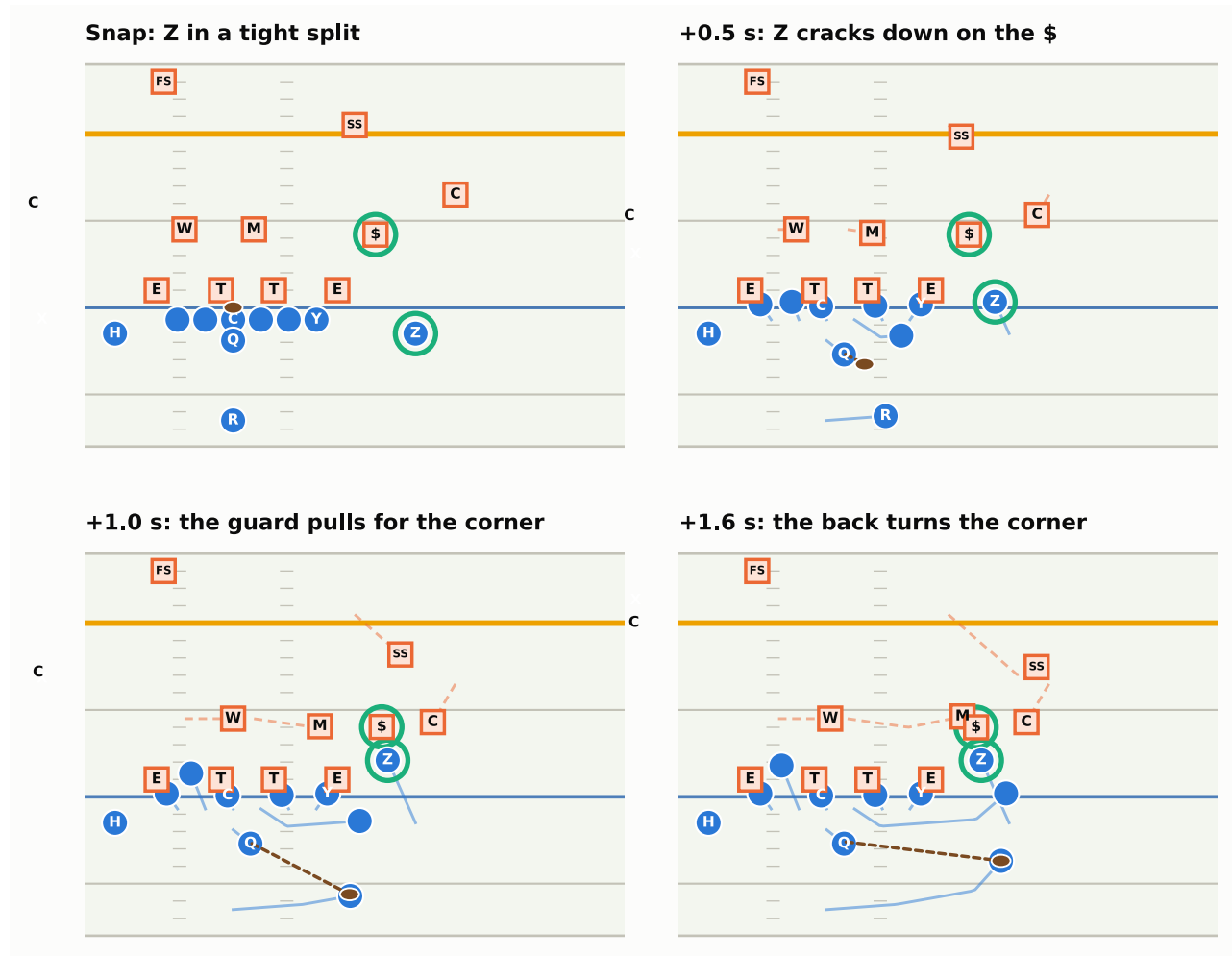


Figure 12: An outside run from a condensed formation, simulated. Z starts in a tight split just outside the tight end. At the snap he blocks down and inside on the nickel back (\$), the defender responsible for the edge, who is watching the backfield and never sees him coming: a crack block. The right guard pulls around to block the cornerback, who has come up to fill the space Z left.

The back takes the toss outside both blocks. From a wide split, Z could never have reached the nickel back in time.

Put those four together and you get the real reason condensed formations took over: **the same alignment threatens a run and a pass**. A receiver in a tight split might be about to crack-block for an outside run, or about to run a crossing route behind a play-action fake, or about to break out into all that space to the sideline. The defense can't tell from his alignment, so it can't cheat. A wide split, by contrast, often announces "pass": a receiver standing twenty yards from the ball isn't going to block anyone in the box.

! Why it exists: condensed splits

Problem: spread formations with wide splits stretch the defense, but they also tell it things. Receivers far from the ball can't block in the run game, and they leave little room toward the sideline. **Answer:** pull receivers in. From a tight split they can crack-block on runs, have room to break either way on passes, and cross each other's paths to create rubs, all from one alignment, so the defense can't guess. **The trade:** condensed formations bring defenders closer to the ball as well. There's less horizontal stretch, more bodies near the box, and more risk of a defender reading the play and getting into the backfield. Offenses that condense usually pair it with motion and play-action to keep the defense moving.

Film room: Rams 2018, condensed 3x1 under Sean McVay

Sean McVay's 2018 Rams went 13-3 and reached Super Bowl LIII.⁵ By our count from nflverse participation data they used **11 personnel on about 93% of their runs and passes**, the most in the league (the median team was about 63%), and they took only **39% of snaps from the shotgun** when the league average was 64%. That combination is the condensed philosophy in numbers: the same three receivers on nearly every snap, often tight to the formation, often with the quarterback under center so play-action looked exactly like the run.

Their receivers, Robert Woods, Cooper Kupp and Brandin Cooks, routinely lined up in reduced splits in 3x1 sets, close enough to crack-block on outside runs and to cross underneath each other on play-action passes. Because the run, the play-action pass and the straight dropback all started from the same alignment, a defense that read the formation learned almost nothing.

What to notice: on a Rams snap from that era, look at how close the outside receiver is to the tight end before the snap. Then watch his first step. On runs he turns *inside* and blocks; on play-action he runs the same first two steps, then releases behind the linebackers who bit on the fake. The Shanahan-McVay family of offenses, and a lot of the league that copied it, still works this way; see the Shanahan tree.

Common misconception

"Spread means pass." Not reliably. Spreading the field pulls defenders out of the box, and that makes runs *easier*, so spread teams run plenty. In the 2016–2022 seasons, when nflverse's participation data still labelled formations by name, plays labelled shotgun (with a back in the backfield) were runs about a quarter of the time. The formation that really does say "pass" is empty, at about 94% (from the backs-in-the-backfield count above). And condensed formations were built precisely so the alignment *doesn't* say run or pass.

How often each formation shows up

The spread family is now the NFL's everyday language, but the most recent seasons have added a twist: heavier personnel and more snaps under center (see Personnel and Classic Formations). The public data doesn't tell us how often teams lined up 2x2 versus 3x1, but it does show the backfield: how many backs stood behind the line, and whether the quarterback was in the shotgun.

⁵Pro Football Reference, 2018 Los Angeles Rams: <https://www.pro-football-reference.com/teams/ram/2018.htm>. Personnel and shotgun shares are this chapter's count from nflverse participation (`offense_personnel`) and play-by-play (`shotgun`), 2018 regular season, runs and passes.

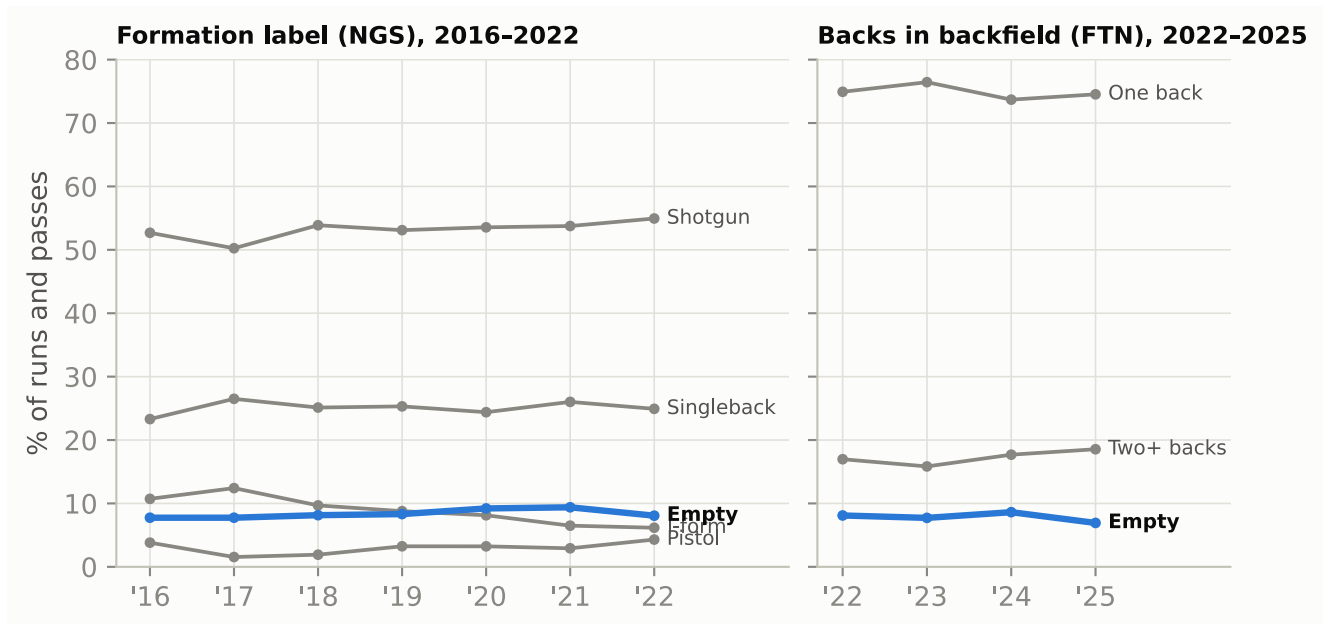


Figure 13: Formations by season, regular-season runs and passes. Left: the formation label in nflverse participation data, 2016–2022 (NGS source), where ‘Shotgun’ means shotgun with a back beside the quarterback and ‘Empty’ is counted separately. Right: FTN charting, 2022–2025, counting backs in the backfield. The two sources use different definitions, so they’re drawn on separate panels; in 2022, the season both cover, they agree that about 8% of plays were empty. Neither source records how the receivers were split (2x2 vs 3x1): that needs tracking data.

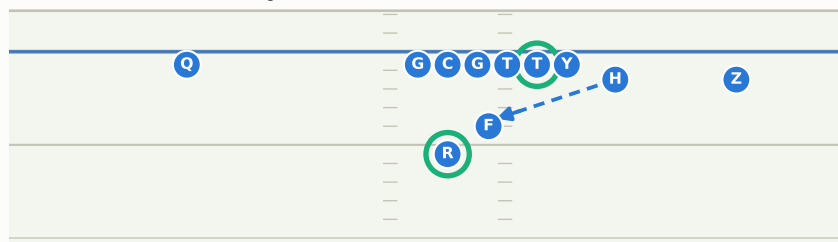
Read the left panel and you’ll see the shotgun was already the NFL’s default look by 2016, with more than half of plays labelled shotgun even before counting empty. Empty itself is a steady seasoning, eight or nine plays in a hundred, every year. The right panel shows the backfield count holding nearly steady from 2022 to 2025: about three plays in four have one back, about one in six have two or more, and empty slipped a little in 2025 as heavier personnel came back into fashion.

The receiver split itself (2x2 against 3x1 against bunch) isn’t in the free play-by-play data. The NFL’s Big Data Bowl tracking files do include it, in a `receiverAlignment` column (values like 2x2, 3x1, 3x2). Once you’ve downloaded a Big Data Bowl season, this cell would build the missing chart:

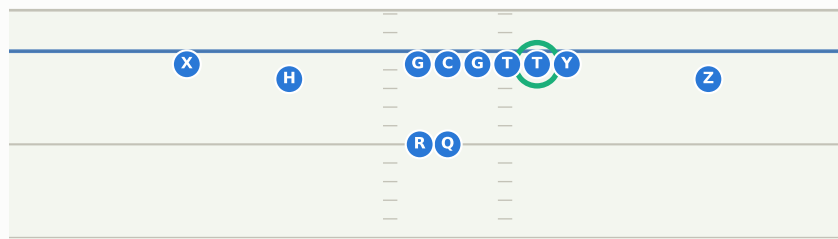
Specialty looks: wildcat, unbalanced, and the swinging gate

Three more formations show up only a few times a season, but when they do, the broadcast usually makes a fuss, and each one is built on a single trick worth understanding.

Wildcat (direct snap to a back)



Unbalanced line (tackle over)



Swinging gate (extra-point try)

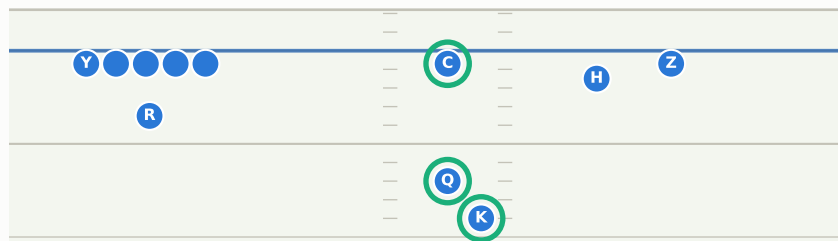


Figure 14: Three specialty looks. Top: a wildcat, drawn after the 2008 Dolphins' version. The ball is snapped straight to a running back (R) in the shotgun spot; the quarterback (Q) is split out wide; a second back (H) is set to run across in motion; and the left tackle has moved to the right side of the line (ringed). Middle: an unbalanced line, 'tackle over'. Four offensive linemen stand right of the center, and the tight end (Y) is still the eligible end beyond them. Bottom: a swinging gate on an extra-point try. Only the center, the holder (Q) and the kicker (K) stay at the ball; most of the line sets up 13 yards to the left with a back behind them. All three are legal: seven players on the line of scrimmage, the eligible players at the ends or in the backfield.

Wildcat

In a **wildcat**, the ball is snapped directly to a running back standing where the shotgun quarterback would be, and the real quarterback is split out wide like a receiver (or is on the sideline). The roots go all the way back to the single wing of the early 1900s, the formation football was built on before the T-formation put a quarterback under center (History).

The logic is arithmetic. On a normal handoff the quarterback gives the ball away and then, for the run game, he's a spectator: the defense can ignore him. The offense has ten players doing something about the run, against eleven defenders. In the wildcat the player who takes the snap is himself a dangerous runner, so the defense has to account for him, and the passer split out wide pulls a defender out to cover him. Add a second back sweeping across in motion, who might take a handoff, and the defense has to account for him too. The extra blocker or runner that this kind of math produces is often called "a hat" (from The Language of the Run Game), and the wildcat is a way to manufacture one.

i Film room: Dolphins 2008, the wildcat in Foxborough

On September 21, 2008, the Miami Dolphins, coming off a 1–15 season and 0–2 to start 2008, visited the Patriots, who hadn't lost a regular-season game since 2006. Miami won 38–13. Running back Ronnie Brown ran for four touchdowns and threw for a fifth, most of it on direct snaps.⁶ In nflverse's play-by-play, six of Miami's snaps that day are described as direct snaps to Brown; they gained 119 yards and four of them ended in touchdowns. The Dolphins went on to finish 11–5 and win the AFC East, and by the same play-by-play text they ran at least 68 direct-snap plays that season, more than the next four teams combined.

The package came from college. Arkansas had used a similar direct-snap set (its "Wild Hog") for Darren McFadden and Felix Jones, and Miami's quarterbacks coach in 2008, David Lee, had been on the Arkansas staff in 2007. Who coined the name "wildcat" is disputed; a high-school coach, Hugh Wyatt, has long said he named his own direct-snap offense that in the 1990s.⁷

What to notice: the Dolphins didn't just snap the ball to a running back. They paired it with an **unbalanced line** (next section) and a second back, Ricky Williams, sweeping across in motion, so the defense faced an extra blocker on one side *and* a runner going the other way. And why did it fade? Defenses caught up. Once they practiced against it, they treated the direct-snap back like a quarterback who can't really throw, kept an extra defender in the box, and the trick lost its surprise. Teams still use it, a few snaps a season, usually near the goal line.

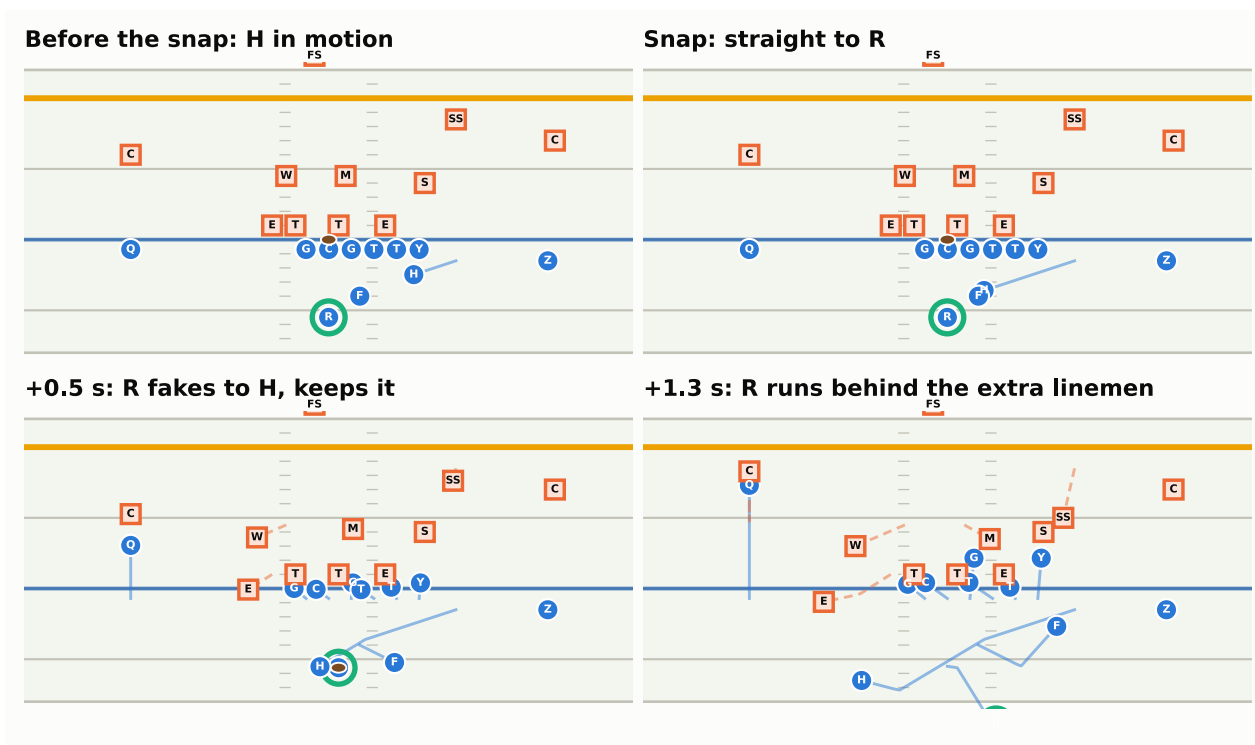


Figure 15: A wildcat run, simulated, after the 2008 Dolphins' version. The second back (H) starts in motion from the right before the snap. The ball is snapped straight to R, who fakes a handoff to H as he runs past and keeps it, running to the right behind the unbalanced line (four linemen and the tight end on that side) with the fullback (F) leading. The left end and the Will linebacker chase the fake to the left. Read the panels in order: before the snap, the snap, the fake, the run.

Unbalanced line (tackle over)

Normally an offensive line is balanced: two linemen on each side of the center. In an **unbalanced line**, one side has more. The usual way is **tackle over**: the left tackle lines up on the right, next to the right tackle, so three linemen stand right of the center

⁶Pro Football Reference box score, Miami Dolphins at New England Patriots, September 21, 2008: <https://www.pro-football-reference.com/boxscores/200809210nwe.htm>. Direct-snap counts are this chapter's text search of nflverse play-by-play descriptions. Season record: <https://www.pro-football-reference.com/teams/mia/2008.htm>.

⁷Wildcat formation, Wikipedia: https://en.wikipedia.org/wiki/Wildcat_formation, which summarizes the competing origin claims (Hugh Wyatt's high-school offense; Arkansas' 2006–07 "Wild Hog" package).

and only one (a guard) on the left. Add the tight end on the right end and the offense has five blockers on the line to the right of the ball.

This is legal as long as there are seven players on the line and the players at each end of the line are the ones the rules allow to be there (a lineman at the end is fine; he just remains ineligible to catch a pass). The rules are in Formation Rules.

Why do it? Defenses align by rules, and most of those rules count linemen and tight ends to figure out which side is strong. An unbalanced line breaks the count. If the defense doesn't notice, or notices late and shifts only partway, the offense has more blockers than the defense has defenders on one side, and it runs there. If the defense does shift a whole gap over, it has had to move several players at once, sometimes leaving a weakness on the other side. And the eligible tight end at the far end can still slip out on a pass, often wide open, because everyone has been taught to watch the run.

Swinging gate

The **swinging gate** is the strangest-looking formation in football. Lined up for an extra point, the offense leaves the center, the holder and the kicker at the ball and sends most of the line, plus a back or two, far off to one side, ten or fifteen yards away from the ball. The name comes from the way the line "swings" out like a gate on a hinge.

The trick is numbers again. The defense has to cover both groups. If it leaves too few players across from the group off to the side, the offense snaps the ball to that group (or tosses it there) and walks in for two points. If the defense matches correctly, the offense usually shifts back into a normal kicking formation and kicks the extra point; the defense has had to scramble to realign and has sometimes left a gap or picked up a penalty doing it. Like the wildcat and the unbalanced line, it tests whether the defense's alignment rules hold up against a picture they rarely see.

Watch for it on Sunday

Watch for it on Sunday: formation and leverage

For one quarter, do this before every offensive snap. It takes about five seconds once you're used to it.

1. **Name the formation.** Is there a back beside the quarterback? (None: empty.) Count receivers left and right of the ball: 2x2, 3x1, 3x2, 4x1. Then add any special word: bunch (three in a cluster), stack (one behind another), nub (a tight end with nobody outside him).
2. **Check the splits.** Are the outside receivers near the numbers, or pulled in tight toward the tight end? Tight splits mean crack blocks, crossing routes and space outside are all possible. A receiver who is unusually tight on one snap is a tell worth remembering.
3. **Do the count.** On each side of the ball, count receivers, then count defenders who could cover them (corners, slot defenders, safeties on that side, a linebacker who has walked out). Say it out loud: "three on two to the right."
4. **Look at the box.** If the defense matched every receiver, how many are left near the ball? Six or fewer against a spread formation means a run has room.
5. **Find the lonely receiver.** In trips or quads, look at the single receiver on the other side. Is there a safety over the top of him, or is he alone with his cornerback?
6. **Check yourself.** After the snap, where did the ball go? Was it to the side where the offense had the numbers, to the isolated receiver, or into a light box?

Predict the play

💡 Predict the play 1: the lonely receiver

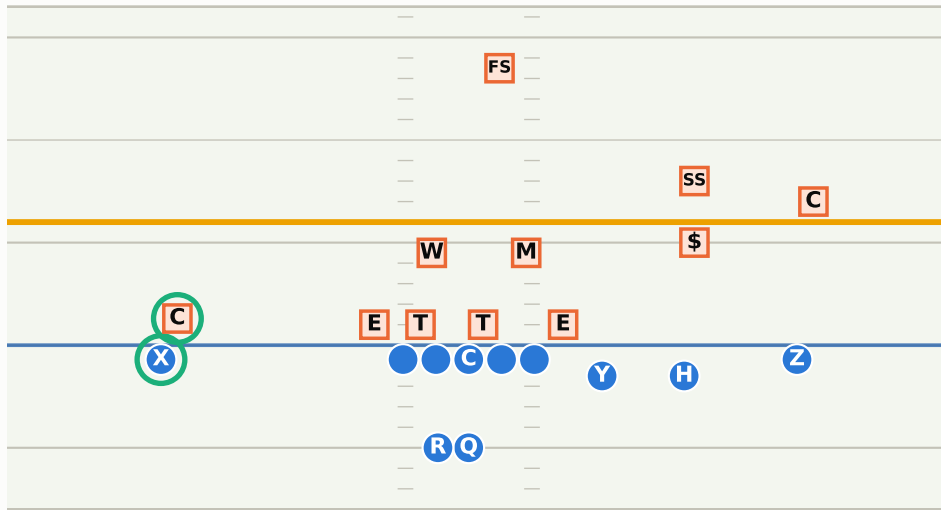


Figure 16: Third-and-6. Trips to the right, the X receiver alone on the left. Before the snap the left safety has drifted toward the middle of the field and the right safety has come down toward the trips side. The left cornerback is pressed up on X, with a one-yard inside shade.

Where is the ball most likely going, and what kind of route?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: six at the line

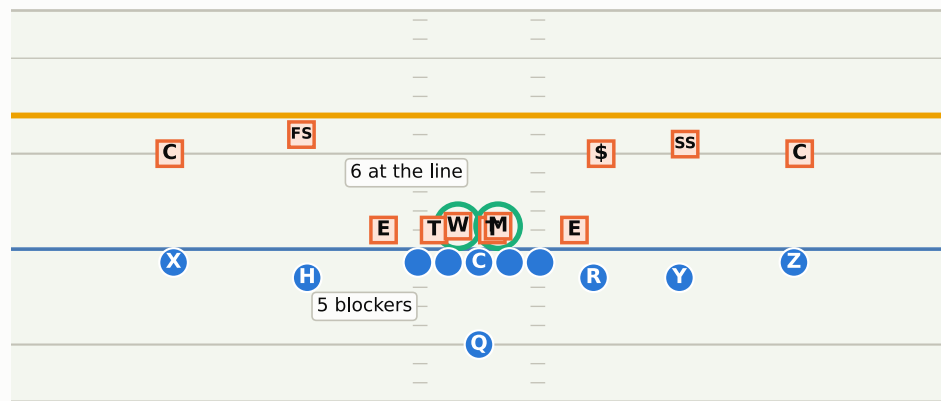


Figure 17: Third-and-7. The offense is in empty, three receivers right and two left. The defense walks both linebackers up to the line of scrimmage beside the four linemen, so six defenders are within a yard of the ball. Every receiver has a defender over him at five to six yards, and nobody is deep.

Count the blockers and the possible rushers. What does the quarterback have to do?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: count the linemen

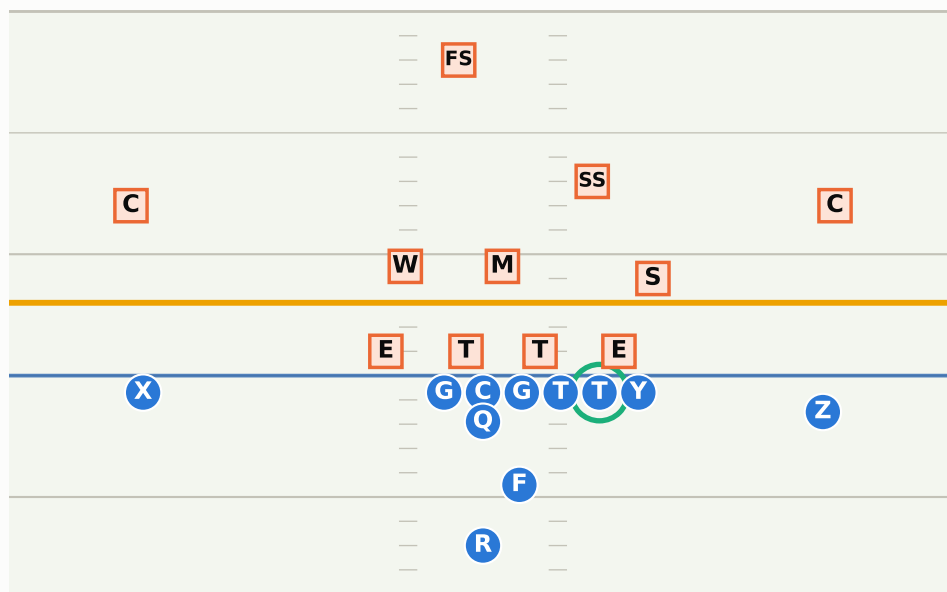


Figure 18: Second-and-3. Look at the offensive line: the left tackle has moved over to the right side, so four offensive linemen stand right of the center, with the tight end beyond them, and only one guard on the left. The defense has lined up exactly as it would against a normal line with the tight end on the right.

Run or pass, and which way?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- Spread formations are named by **counting receivers** on each side of the ball (backs in the backfield don't count): **2x2** (doubles), **3x1** (trips), **3x2** (empty), **4x1** (quads), plus **bunch**, **stack** and **nub** for how receivers are grouped.
- **Width does two jobs.** It pulls defenders out of the box, which helps the run game, and it forces the defense to align to receivers before the snap, which shows the quarterback the coverage. The price is fewer blockers near the ball.
- **Trips and quads overload one side.** The defense either rotates help over, leaving the lone receiver one-on-one, or stays balanced and is outnumbered or thin in the box. Watch the safeties to see which it chose.
- **Empty is the strongest pass tell** in the formation family (about 94% passes, 2022–2025), and its cost is protection: five blockers, so the quarterback must get rid of it fast or be a run threat himself.
- **Condensed splits** bring receivers close to the ball so one alignment can threaten a crack block, a two-way go, rubs and room to the sideline, which hides run from pass.
- **Specialty looks** (wildcat, unbalanced line, swinging gate) each test whether the defense's alignment rules can count. When the defense miscounts, the offense goes where the numbers are.

Connections

Builds on: Formation Rules and the Language of Alignment (seven on the line, receiver labels, splits and the numbers), Personnel (who is on the field before the formation is chosen) and Classic Formations (the tight, two-back world the spread grew out of).

Sets up: A First Look at Defense is the other side of the count: the packages, the box and the one-high and two-high shells you've been counting against. Motion and Shifts as Weapons turns these still pictures into moving ones, changing the formation before the snap to make the defense reveal itself again. Further on: Passing Game Fundamentals, The Quick Game (leverage and quick throws to the side with the numbers), Dropback Concepts (rub routes from bunch), Perimeter Runs and Misdirection (crack blocks), and Disguise, Rotation, and Split-Field Coverage (how defenses handle trips without leaving X alone). The history is in The College Laboratory and The Modern Era; the condensed, under-center version lives in the Shanahan tree.

i Go deeper

- Chris B. Brown, *The Art of Smart Football* (2015) and *The Essential Smart Football* (2012): short essays on the spread, the run and shoot, the Air Raid and why formations work.
- Tim Layden, *Blood, Sweat and Chalk* (2010): the coaches behind the run and shoot, the spread option and the wildcat, and the problems they were solving.
- S. C. Gwynne, *The Perfect Pass* (2016): Hal Mumme, Mike Leach and the birth of the Air Raid.
- Glenn “Tiger” Ellison, *Run and Shoot Football: Offense of the Future* (1965), the original pitch for spreading the field, still readable.

Answers to Predict the play

Answer 1.

To X, one-on-one, probably outside: a fade or a back-shoulder throw (or a quick slant if the corner overplays outside).

Do the count. On the trips side the defense has four players (corner, nickel, a safety walking down, and the Mike leaning that way) against three receivers, plus the other safety drifting toward the middle. On the left, X has a pressing cornerback and nobody deep on his side of the field. The corner’s inside shade takes away the slant and dares X to win outside along the sideline, which is exactly where a big receiver with a two-yard head start wants to go.

Two cautions. First, the offense may not trust the picture: defenses disguise rotations, and the “rotated” safety could spin back at the snap (Disguise, Rotation, and Split-Field Coverage). Second, on third-and-6 the offense may have called a trips-side play anyway; the quarterback’s job is to confirm X’s one-on-one after the snap before throwing. But when you see trips, a rotated safety and a pressed X, X is the first place to look.

Answer 2.

Get the ball out fast, to the receiver who wins quickest, probably on a short in-breaking route. Empty leaves five blockers. If all six defenders at the line rush, one is unblocked the moment the ball is snapped, and he’ll reach the quarterback in about two seconds. With no deep safety, every receiver is in man coverage, so the quarterback is looking for a quick slant, a hitch or a “hot” throw to whoever has the most room, before that free rusher arrives. If he gets it out on time, there’s nobody deep to stop a receiver who makes the first man miss: this is how a short throw turns into a long touchdown against a blitz.

The defense might also be bluffing. Two linebackers on the line can drop back into coverage at the snap, so the quarterback who panics and throws hot into a defender has been fooled. That “show pressure, then drop” idea is Zone Blitzes and Simulated Pressure.

Answer 3.

A run to the right, at the unbalanced side. Count the blockers to the right of the center: the right guard, the right tackle, the tackle who moved over, and the tight end, four, with a fullback who can lead through the hole as a fifth. Now count the defenders on that side who are close enough to stop a run: the defensive tackle, the defensive end, the Sam linebacker, and perhaps the Mike, with the strong safety eight yards back. The defense aligned to the tight end, as its rules say, but its rules didn’t count the extra tackle. That’s a numbers advantage at the point of attack on a short-yardage down.

What would change the answer? If the defense had shifted its whole front one gap toward the unbalanced side, the advantage disappears, and the offense might run the other way (where only a guard and the center are left, but the defense has moved away too) or fake the run and throw to the tight end. The tell is whether the defense moved. Watch for it the next time a broadcast mentions “an unbalanced look”: the first thing the defense’s middle linebacker does is point and shout, trying to get everyone to slide over before the snap.